

–| Linear Regression Industry Demonstration |--

This session is to be scheduled on the weekend after the launch of the Linear Regression module. The students are expected to have completed >50% of the module and at a minimum, be familiar with the topic. This session is 1.5 hours long.

Learning Objective:

In this session, we wish to cover an interesting industry problem which can be solved using multiple linear regression.

- Understanding the business problem and the data.
- Building the simple regression model
- Model evaluation - Choosing an appropriate threshold
- Making predictions.

Prerequisite(s):

First 3 sessions of the Linear Regression module - Simple and Multiple Linear Regression, Implementing linear regression in Python

Brief Agenda:

1. Part-I: Model Building (40 mins)
 - Introduction - (5 mins)
 - Understanding the business problem and the data - (5 mins)
 - Data Exploration, Cleaning, and Visualisation - (15 mins)
 - Model building - (15 mins)
2. Part-II: Model evaluation (30 mins)
 - Understanding model evaluation metrics- (10 mins)
 - Model improvement - (20 mins)
3. Part-III: Model Predictions (10 mins)
4. Part-III: Doubt Resolution (10 mins)

Detailed Lesson Plan

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Part-I: Model Building (40 mins)	Introducti onand concepts brushup (5 mins)	The instructor welcomes the learners 5 and gives a brief agenda for this session Instructor asks the students about their journey in ML and addresses a couple of common problems that everyone is facing in the modules.	3	Social Support
	Understa ndi ng the problem statement and the data(5 mins)	The learners will be provided the dataset 5 at the start of the session so they can try exploring it themselves and follow along if they wish to. We are also sharing some sample, real-world datasets like Boston Housing prediction or Used car price prediction for learners who wish to work on other datasets independently. The instructor explains the problem statement and gives the necessary background information, the data source etc. Mention in brief the processes involved.	-	Hands-On and Learner Tasks
	Data explorati on, cleaning and visualisat io n (15 mins)	Perform the basic EDA and the 15 visualisations on the data including, if necessary: • Identifying erroneous data • Null imputations • Dummy variables • Dealing with outliers • Scaling etc.	3	Hands-On and Learner Tasks

		Make inferences on the data.		
	Model Building (15 mins)	Perform the following operations 15 • Train test split • Train first model using all features	-	Hands-On and Learner Tasks
Part-II: Model Evaluation (30 mins)	Understanding evaluation metrics (10 mins)	Give some context on the different metrics used to evaluate linear regression model: 10 • Mean Absolute Error(MAE) • Mean Squared Error(MSE) • RMSE • R squared • Adjusted R Squares	2	Hands-On and Learner Tasks
	Model improvement (20 mins)	• Use RFE to select significant features • Train model until a good level of accuracy is achieved • Perform Residual Analysis till error terms attain normal distribution 20	2	Hands-On and Learner Tasks
Part-III Model Predictions (10 mins)	Making model predictions	Make predictions with the final set of variables. 10 Check correlation between final predictor variables.		Hands-On and Learner Tasks
Part-IV: Doubt Resolution	Doubt resolution	Address some common doubts related to linear regression and the industry example demonstrated. 10	-	Doubt Resolution

on (10 mins)				
Total (in mins)		~90		